

Eric Gitangu

Software Engineering Lead / Architect — Distributed Systems and AI/ML

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PROFESSIONAL SUMMARY

Software Engineering Lead and Architect with 10+ years shipping event-driven distributed systems across healthcare ML, fintech, energy access, and government-scale infrastructure. Deep expertise in CQRS, event sourcing, hexagonal/ports-and-adapters, and saga orchestration — applied at Vishnu Systems (ML security for SiMD/SaMD medical devices), Ignite/ENGIE Energy Access (PAYG solar across 7 African markets), Baw Bab Technologies (digital addressing for 2.3M Nairobi residents), and Unicorns (multi-tenant SaaS super-app). Polyglot across Python, Java, Rust, TypeScript, Go, Kotlin, Swift, and Clojure. Founded Deveric mentorship initiative (15+ engineers mentored) and hold 80+ validated technical certifications.

EXPERIENCE

Team Lead — QA & Engineering (SDET & Platform) — Ignite Energy Access (formerly ENGIE Energy Access) March 2022 - Present
Nairobi, Kenya

Leading QA and engineering automation for PAYG solar energy infrastructure serving communities across 7 African markets. Driving platform reliability, observability, and test automation strategy.

- Architected and led migration of legacy manual QA workflows to a fully automated SDET pipeline using Playwright, Cypress, and custom Ruby harnesses — reducing regression cycle time by 60%
- Designed and deployed observability stack (OpenTelemetry, Prometheus, Grafana, Jaeger, ELK) across 7 African market deployments, cutting mean-time-to-detection from hours to minutes
- Built and maintained CI/CD pipelines integrating automated security scanning, performance benchmarks, and contract testing across microservices
- Drove cross-functional alignment between product, engineering, and data teams to establish quality gates and SLA-driven release criteria
- Introduced chaos engineering practices and load testing frameworks (K6, Locust) to validate system resilience under peak PAYG billing cycles
- Designed shared RabbitMQ-backed bounded contexts (Payment, Token, Subscription) as a reusable platform layer serving all product teams; decoupled synchronous PAYG processing, eliminated peak-load timeouts, unblocked release cadence from 2 to 12+ deploys per week — a paved-roads / golden-paths platform pattern adopted across the engineering org
- Delivered 900+ end-to-end automated scenarios across payments, IoT provisioning, token lifecycle, and subscription flows; reduced regression from 3 days to 45 minutes; enabled continuous deployment across 7 markets
- Line-managed 4 engineers across Kenya, Uganda, and Tanzania — 2 promoted to senior within tenure; enforced architectural standards and OWASP Top 10 secure coding through structured code reviews

Python Ruby Node.js PostgreSQL Redis Kafka RabbitMQ AWS OpenTelemetry Playwright Cypress ELK Prometheus
Grafana Jaeger Redshift Kubernetes K6 Locust

Director of Engineering / Sole Technical Founder — Vishnu Systems, Inc.

Jan 2024 – Aug 2025

Atlanta, GA

Founded and built an ML-driven security platform for Software-in-a-Medical-Device (SiMD) and Software-as-a-Medical-Device (SaMD) applications, targeting HIPAA-compliant vulnerability detection and compliance automation for healthcare providers.

- Designed and built a FastAPI orchestration layer with CQRS and event-sourced domain model, routing scanning jobs through async task queues with saga-based compensation for partial failures
- Implemented high-performance Rust+PyO3 modules for binary analysis and SBOM generation, achieving 50-100x throughput improvement over pure Python equivalents
- Built ML pipelines combining CNNs for medical image metadata analysis, RAG for compliance document retrieval, and GNNs for dependency-graph vulnerability propagation
- Developed compliance automation tooling mapping scan results to HIPAA, FDA 21 CFR Part 11, and IEC 62304 requirements with automated evidence generation
- Delivered full MLOps stack: Airflow orchestration, MLflow experiment tracking, SageMaker/Vertex AI training, and A/B model deployment with canary rollouts
- Implemented Circuit Breaker + exponential backoff in Go to absorb burst load on GPU evaluation workers — custom half-open state machine equivalent of Resilience4j (5s initial interval, 2x multiplier, 60s ceiling) with async Kafka drain on open-circuit; zero customer-facing timeouts during a live degradation event, circuit self-healed without manual intervention
- Productionised AI-assisted development tooling — integrated LLM APIs (OpenAI, Anthropic Claude) for automated compliance documentation generation, reducing manual report assembly to zero; foundation for AI-assisted DevOps tooling experimentation
- Shipped idempotent SQS / SNS / Kinesis + Kafka pipelines with DLQs and exactly-once semantics; full CloudWatch + X-Ray observability and customer-facing SLAs

Python FastAPI Rust PyO3 Go React Node.js TensorFlow PyTorch Airflow MLflow AWS Lambda DynamoDB Kafka
Kubernetes Docker OpenAI Anthropic Claude SQS SNS Kinesis CloudWatch X-Ray

Lead Software Engineer — Baw Bab Technologies, Inc.

Jan 2023 – Aug 2025

Seattle, WA / Nairobi, Kenya

Led engineering for a government-scale digital addressing and logistics platform, enabling formal address assignment for previously unaddressed urban areas in Nairobi, Kenya.

- Led a 5-engineer team delivering a digital address platform that assigned formal addresses to 2.3 million Nairobi residents, integrating with Kenya's national addressing initiative
- Architected computer-vision and GIS pipeline using Google Cloud Vision for address-plate recognition, PostGIS for spatial queries, and Google Cloud Pub/Sub for event-driven geocoding workflows
- Built full-stack delivery: React/Next.js dashboard for administrators, Express API gateway, MongoDB document store, and GCP Cloud Run deployment with auto-scaling
- Applied advanced geospatial algorithms (Dijkstra's, A*) for last-mile route optimization, reducing delivery time by 35% in pilot zones
- Initial production deployment was on Azure (App Service, Azure SQL Database, Azure Blob Storage, Azure DevOps CI/CD pipelines, with Azure Virtual Machines for runtime management during the on-premises-equivalent phase) before evolving to a hybrid Azure + GCP Cloud Run model — a direct precedent for predominantly on-prem Windows Server environments evolving toward hybrid cloud
- Designed a Backend-for-Frontend (BFF) platform layer in Go that separated field-agent mobile from county-government web surfaces; defined OpenAPI specs, RESTful APIs, and Kafka event pipelines as shared platform standards adopted by Nairobi County Government as official infrastructure

React Next.js Node.js Express Go Python Kafka MongoDB Google Cloud Vision Google Cloud Pub/Sub GCP Cloud Run PostGIS
Docker Kubernetes Azure App Service Azure SQL Database Azure Blob Storage Azure DevOps Azure Virtual Machines OpenAPI

Senior Software Engineer — Reinsurance Group of America (RGA)

Jan 2022 – Dec 2023

Remote

Modernized legacy actuarial and billing systems into microservices architecture, supporting reinsurance treaty management, premium calculation, and claims processing workflows.

- Decomposed monolithic Java/Spring Boot application into 12 bounded-context microservices, defining clear service boundaries using DDD aggregate patterns
- Led polyglot modernization: Java 17 for core actuarial engines, Groovy/Grails for rapid internal tooling, Node.js for API gateway and BFF layers
- Built React frontend for treaty management dashboard with real-time premium calculation, replacing legacy JSP/Struts interface
- Identified and resolved critical performance bottlenecks in batch premium calculation pipelines, achieving 4x throughput improvement through query optimization and caching strategies
- Established service boundary contracts using OpenAPI specs and Pact contract testing, enabling independent deployment across 6 engineering teams

Java Spring Boot Groovy Grails React Node.js AWS ECS PostgreSQL MongoDB Redis

Founder & Principal Consultant — Deveric Technologies (Freelance)

Jan 2019 – Feb 2022

Nairobi, Kenya / Remote

Founded consultancy and mentorship initiative, delivering production systems for local businesses while mentoring 15+ engineers (6+ to advanced levels) in software architecture, DevOps, and system design.

- Founded Deveric mentorship initiative (2019–present): 15+ engineers mentored across career stages, 6+ advanced to senior/lead roles through structured curriculum covering system design, algorithms, and production engineering
- Delivered 10+ production projects across fintech, e-commerce, and logistics for Kenyan SMBs using MEAN/MERN stack with REST/GraphQL APIs and Redis caching
- Mentees collectively earned 80+ validated technical certifications across AWS, Azure, GCP, AI/ML, and blockchain
- Architected payment platform integrations with M-Pesa, multiple PSPs, and multi-currency support for cross-border transactions

React Node.js Go Rust FastAPI MongoDB PostgreSQL Redis AWS GCP

Software Engineer — Vecima Networks Inc.

Jan 2018 – Jan 2019

Peachtree City, GA

Contributed to petabyte-scale video streaming and telecommunications network management infrastructure, working on low-level systems for content delivery and OSS/BSS workflows.

- Developed low-level C and assembly modules for video packet processing in petabyte-scale streaming infrastructure, optimizing for throughput and minimal latency
- Built reactive stream processing pipelines for real-time network telemetry, enabling sub-second anomaly detection across distributed CDN nodes
- Designed fault-tolerant architecture patterns for content delivery, implementing circuit breakers and retry mechanisms that maintained 99.95% uptime during peak traffic

C Assembly Java Python PostgreSQL RabbitMQ Kubernetes

Software Engineer — Veracode

Dec 2015 – Jan 2018

Burlington, MA

Built security scanning platform features, contributing to the decomposition of a J2EE monolith into microservices while developing CI/CD integrations for enterprise DevSecOps workflows.

- Designed and implemented HATEOAS-compliant REST APIs using Spring Boot, JPA/Hibernate, and C#/.NET, supporting multi-tenant scan orchestration with role-based access control
- Built CI/CD DevSecOps integrations (Jenkins, GitLab CI, Azure DevOps) enabling customers to embed security scanning directly into their build pipelines
- Contributed to OWASP Top 10 and NIST SSDF compliance features, implementing automated policy engines that mapped scan findings to regulatory requirements
- Owned the Veracode Visual Studio plugin (.NET 4.5 / Windows platform component) end-to-end — enabling developers to run SAST analysis at file, project, solution, or pipeline granularity directly inside Visual Studio, or to inject the .NET backend into enterprise CI/CD systems for shift-left secure coding. The entire platform stack ran on-premises on Windows Server (2012/2016 era), the most direct precedent in this resume to predominantly-on-prem Windows Server platforms
- Led the platform upgrade to enforce TLS 1.2 across the .NET 4.5 plugin and on-premises Windows Server backend services — a compliance-critical delivery for enterprise customers in regulated environments (financial services, healthcare, government), requiring coordinated changes across the plugin, client-facing APIs, and on-prem server configuration
- Owned the full SDLC and continuous-delivery pipeline for plugin releases — produced MSIX packaging and signed delivery artefacts, managed versioning and backward compatibility for clients on mixed .NET Framework versions, and rollout sequencing across a large installed base of Fortune 500 customers running on-prem Windows Server

Java Spring Boot JPA Hibernate C# .NET .NET 4.5 ASP.NET Visual Studio SDK MSIX Windows Server 2012/2016 Python

Docker Kubernetes Jenkins GitLab Azure DevOps OWASP Top 10 NIST SSDF

Software Engineer (Co-op) — IBM

Sep 2013 – May 2014

Littleton, MA

Contributed to IBM Web Experience Factory (formerly WebSphere Portlet Factory) test automation and build pipeline optimization, establishing test-automation-first discipline.

- Built automated regression test suites for IBM Web Experience Factory, establishing test-automation-first discipline that reduced manual QA cycles by 40% and accelerated release schedules
- Optimized Ant/Maven build pipelines with parallelized test execution and incremental compilation, cutting build times by 30%

Java Ant Maven Jenkins

SELECTED PROJECTS

PawaCloud Assistant — AI Cloud Advisor · pawacloud-web.fly.dev

AI chatbot with Gemini 2.5 Flash, Rust+PyO3 text processing, SSE streaming, Google OAuth, and PostgreSQL persistence on GCP Cloud Run

Python FastAPI Rust PyO3 Gemini AI Next.js TailwindCSS PostgreSQL Google Cloud Cloud Run

UniRides — Ride-Sharing Reimagined · unirides.unicorns.run

Production-grade ride-sharing platform with 5 Google Maps APIs, real-time tracking, Kenyan market-rate pricing, and FastAPI backend on GCP Cloud Run

Next.js TypeScript TailwindCSS Google Maps Platform FastAPI Python SQLAlchemy PostgreSQL Redis WebSocket

LendStream · nanolend-app.vercel.app

Polyglot micro-lending platform — Java, Clojure, React, Kafka, CQRS

Java 21 Spring Boot Clojure React TypeScript Kafka QStash PostgreSQL Redis Docker

Vunachain — AgriTech Contract Farming & Traceability · vunachain.com

Blockchain-backed contract farming and agricultural supply chain traceability platform on AWS serverless with Rust Lambda functions

Rust AWS Lambda ARM64/Graviton EventBridge SQS DLQs S3 RDS PostgreSQL CloudWatch X-Ray

Wave Application — ML Engineer · wave-apply.ericgitangu.com

Full-stack polyglot submission for Wave's Senior ML Engineer role with Next.js 16 PWA, Rust+PyO3 Lambda backend, and AWS CDK infrastructure

Next.js Rust Python AWS CDK Bedrock SageMaker Upstash Redis Fly.io TypeScript

Secure Messenger Desktop · secure-messenger-desktop.fly.dev

Electron + React + TypeScript secure messenger with real-time WebSocket sync, AES-256-GCM encryption, SQLite storage, and Prometheus/Grafana observability

Electron React TypeScript SQLite Redux Toolkit WebSocket AES-256-GCM Prometheus Fly.io

UniCorns - Multi-Tenant SaaS Super-App · unicorns.ericgitangu.com

Comprehensive multi-tenant business platform for African SMBs with 5 Rust microservices, healthcare EMR, M-Pesa payments, and AI features

Rust AWS Lambda DynamoDB Next.js Kotlin Claude AI M-Pesa

Interactive Resume with AI Chatbot · resume.ericgitangu.com

Modern interactive resume with AI-powered chatbot, career visualizations, and downloadable formats

Next.js TypeScript AI/LLM Recharts Tailwind CSS Fly.io

CORE TECHNICAL SKILLS

Architecture & Systems

Microservices · Event-Driven · CQRS · Event Sourcing · Saga Orchestration · Hexagonal / Ports & Adapters · DDD · System Design · Multi-tenant · API Design · BFF Pattern · Railway-Oriented Programming · Actor Model · CRDTs · Transactional Outbox · Circuit Breakers

Backend & APIs

Python · Node.js · Java · Rust · Go · Scala · Clojure · C/C++ · Django · FastAPI · Groovy/Grails · Axum · GraphQL · REST APIs

AI/ML

TensorFlow · PyTorch · LLMs · OpenAI · Anthropic Claude · Google Gemini · RAG Systems · CNNs · GNNs · Transformers · Apache Airflow · MLflow · Vertex AI · pgvector

Databases & Data

PostgreSQL · MongoDB · Redis · DynamoDB · MySQL · Kafka · RabbitMQ · Redshift · ETL/ELT Pipelines

Cloud & DevOps

AWS · Kubernetes · Docker · GCP · Azure · CI/CD · Terraform

Frontend & Mobile

React · Next.js · TypeScript · Angular · Vue.js · Tailwind CSS · PWA · Kotlin + Jetpack Compose · Swift/SwiftUI · Electron

Secure APIs & Compliance

OWASP Top 10 · NIST SSDF · HIPAA · FHIR/HL7 · AES-256-GCM · OAuth2/JWT

Observability

OpenTelemetry · Prometheus · Grafana · Jaeger · ELK Stack · AWS X-Ray · CloudWatch

Web3 & Blockchain

Solidity · DApps · Web3.js

.NET / Microsoft Platform

.NET 10 / .NET 8+ · .NET 4.5 / Windows Server (legacy) · ASP.NET Core · C# · Entity Framework Core · MediatR / CQRS · Polly (retry / circuit breaker) · FluentValidation · xUnit / FluentAssertions / Moq · Playwright .NET · Serilog · Scalar / Swagger OpenAPI · Visual Studio plugin development · MSIX packaging / signed delivery · IIS / Windows Server hosting · Azure App Service · Azure SQL Database · Azure Blob Storage · Azure Key Vault · Azure Functions · Azure DevOps Pipelines · Azure Virtual Machines · Bicep IaC

Domains

Healthcare ML · Digital Health (SiMD/SaMD) · Fintech · Energy Access (PAYG Solar) · AgriTech · Multi-tenant SaaS · Government-scale Infrastructure

EDUCATION

Bachelor of Science in Computer Science · Minor: Mathematics — University of Massachusetts at Lowell

Sep 2011 – May 2015

- Microsoft Scholarship Recipient (2012-2013) — Awarded for academic excellence and demonstrated potential in software engineering
- Google Scholarship Recipient (2011-2012) — Awarded for outstanding achievement in computer science coursework and research
- Dean's List: Fall 2010, Spring 2011, Fall 2011, Spring 2012, Spring 2013, Fall 2013, Spring 2014, Fall 2014, Spring 2015
- Omicron Delta Kappa (ODK) Honor Society

CERTIFICATIONS & OPEN SOURCE

- **Open Source Contributions:** 20+ production-ready public repositories on GitHub across Rust, Go, TypeScript, Python, and Java — spanning distributed systems, ML pipelines, fintech, and developer tooling
- **Mentorship & Teaching:** Founded Deveric mentorship initiative (2019–present): 15+ engineers mentored, 6+ advanced to senior/lead roles. CodePath.org Volunteer Student Teacher (May–Aug 2024): led Advanced TIP103 cohort in algorithms and data structures. Technical writing at developer.ericgitangu.com.
- **Certifications:** 80+ validated technical certifications across AWS, Azure, GCP, AI/ML, blockchain (Solidity/Polygon/Stellar), and application security — demonstrating continuous learning across the full stack.